

Original papers

Calibration-enhanced multi-view RGB-D vision for robust recognition and 3D localization of strawberries under occlusions[☆]Shimin Hu^{a,b}, Meili Sun^a, Chunjiang Zhao^{a,b}, Ya Xiong^a^{*,*}^a Intelligent Equipment Research Center, Beijing Academy of Agriculture and Forestry Sciences, Beijing, 100097, China^b School of Intelligence Science and Technology, University of Science and Technology Beijing, Beijing, 100083, China

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ABSTRACT

Accurate 3D localization remains a bottleneck for strawberry-harvesting robots due to occlusions, calibration errors, and ineffective multi-view fusion. This paper presents a comprehensive solution integrating a two-sided checkerboard calibration with residual modeling (TCC-RM) method for high-precision extrinsic calibration of dual RGB-D cameras and hand-eye systems, alongside a multi-RGB-D fusion pipeline for robust strawberry perception under occlusions. The TCC-RM method combined rigid initialization with sphere-based residual compensation to address nonlinear distortions. For multi-view perception, we introduced a modified 3D Intersection over Union (IoU) metric to enhance occlusion-resistant strawberry matching and compare the sequence of coordinate transform among multiple RGB-D frames and the base frame. Experimental results demonstrated that the TCC-RM reduced the calibration errors (RMSE) to 7.23 mm (dual-RGB-D) and 6.74 mm (hand-eye), achieving 49.23% and 69.3% improvements over the traditional TCC method, respectively. Compared with single-RGB-D configuration, the multi-RGB-D fusion reduced the Euclidean localization errors by 2.84%–39.21% and expanded the perception range by 42.86% (mild occlusion). The strawberry perception pipeline achieved RMSEs of 6.31 mm (mild occlusion) and 12.62 mm (moderate occlusion), representing minimum improvements of 11.7% over any single-view configuration. The framework, validated online on the HarvestFlex platform, achieved high-precision 3D localization in greenhouse environments, thereby enhancing the deployment readiness of strawberry-harvesting robots.

1. Introduction

As global agriculture advances toward intelligent and unmanned operations, fruit and vegetable harvesting robots have emerged as a key technology to improve efficiency and ensure food security. Strawberries, as delicate and high-value horticultural crops, are a major focus due to their short ripening period, limited shelf life, and susceptibility to damage (Parsa et al., 2023). Perceiving and localizing strawberries constitute the initial and most critical phases of automated harvesting. However, the complexity of the strawberry harvesting environment introduces considerable challenges to the reliability and accuracy of these processes. Dense clustering, occlusions from leaves and neighboring fruits, and irregular structures hinder the stable 3D localization (Fu et al., 2020; Tituaña et al., 2024).

To address the challenge of occlusion in strawberry localization, researchers have explored a variety of strategies. Ge et al. combined

density clustering with spatial priors to estimate positions, achieving 74.1% picking success in tabletop setups but with poor generalization to natural fields (Ge et al., 2019). Yu et al. introduced rotated bounding boxes in R-YOLO for embedded posture prediction, yet the handling of occlusion remained limited (Yu et al., 2020). LiY et al. incorporated RGB-D fusion and centroid regression in YOLOv7 to enhance Z-axis accuracy, though performance relied heavily on detection quality (Li et al., 2024a). To enhance occlusion adaptation, LiT et al. employed 2D segmentation and point cloud cropping for centroid estimation, improving accuracy but vulnerable to segmentation errors (Li et al., 2022). Gong et al. fused RGB, depth, and infrared data, achieving 2 mm error under mild occlusion, but with slow processing (456 ms/frame) (Gong et al., 2022). Kang et al.'s OccluInst used attention-guided RGB-D fusion for occlusion-aware recognition, though its effectiveness for strawberries remains unverified (Kang et al., 2025). Multi-camera systems

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offer greater resilience to occlusion. Mejia et al. used stereo vision on mobile platforms for spatial mapping, improving recognition by 14%, but lacked fine localization (Mejia et al., 2023). Zheng et al. applied multi-view instance matching for large-scale occluded target merging, but resolution was insufficient for close-range detection (Zheng et al., 2023).

To achieve high-precision 3D localization of strawberry targets, multi-camera systems are widely adopted due to their complementary viewpoints. However, their performance largely depends on the accuracy of inter-camera extrinsic parameters, making calibration a central research focus. The classic checkerboard method (Zhang, 2000) is effective in static or close-range setups but relies on overlapping fields of view, limiting its use in wide-baseline systems. To address this, researchers have proposed dual-sphere targets (Wei et al., 2016) and auxiliary high-precision cameras (Sun et al., 2016), which improve flexibility but increase complexity and cost. Optimization-based methods such as baseline correction (Darwish et al., 2019), multi-temporal adjustment (Kumar et al., 2018), moving-sphere trajectories (Su et al., 2018), and robust circle detection (Kwon et al., 2019) enhance calibration under challenging conditions but often lack real-time efficiency. To improve robustness and adaptability, Liu et al. (2022) introduced a circular calibration board with sub-pixel edge extraction. Yuan et al. (2025) proposed an evaluation strategy for non-overlapping views based on residual analysis, while Xu et al. (2021) used person re-identification to establish correspondences across wide-baseline views without dedicated targets. Hou et al. (2025) developed a hierarchical framework combining multiple modalities, though with high system complexity. In recent years, deep learning techniques have been widely applied in multi-camera system calibration. For instance, Duan et al. proposed a deep neural network-based method that achieves high-precision calibration using small calibration boards in large field-of-view setups, improving efficiency and accuracy (Duan et al., 2024). Additionally, Talakoob introduced a real-time neural network recalibration method for multi-camera systems, dynamically adjusting camera parameters in real-time based on changing poses, ensuring system robustness (Talakoob, 2024). However, their reliance on substantial data and computational resources, coupled with challenges in interpretability and generalization, raises concerns about practical deployment in resource-limited settings.

Hand-eye calibration is essential for establishing spatial transformations between vision systems and robotic manipulators, enabling accurate execution based on visual perception. The classical $AX=XB$ formulation (Tsai et al., 1989) offers efficient computation but is sensitive to pose noise and large viewpoint changes. To improve robustness, Wu et al. (2023) used a single-sphere point cloud model combined with quaternion and Kronecker product methods to jointly solve rotation and translation, though performance depends heavily on point cloud quality. Shin et al. (2024) enhanced observability through pure rotational motions around orthogonal axes, requiring precise motion control. Cui et al. (2024) proposed a two-stage closed-form method with RANSAC and error refinement, achieving higher accuracy under noisy conditions but increasing computational burden. Wang et al. (2025) introduced pose graph optimization to jointly estimate transformations and system errors, improving global consistency but reducing real-time feasibility. Geometric constraint methods, such as those by Yang et al. (2022) and Zhong et al. (2025), simplify deployment but are limited by joint precision or depth perception. Recent intelligent approaches, including neural network modeling with evolutionary optimization (Zhu et al., 2022), and target-free self-calibration via point cloud matching (Li et al., 2024b), improve autonomy but rely on high data quality. Multi-sensor fusion strategies (Gautam et al., 2025) enhance adaptability but impose strict synchronization requirements. Recent advancements, such as the methods proposed by Valassakis et al. (2021) and Tang et al. (2025), have significantly improved calibration processes. Valassakis et al. explored deep learning for eye-in-hand camera calibration from a single image, showcasing the potential of direct regression models to

outperform classical methods in noisy environments (Valassakis et al., 2021). Tang et al. proposed Kalib, a markerless hand-eye calibration method that uses reference point tracking, forward kinematics, and a PnP solver to estimate the camera-robot transformation, offering a more user-friendly, low-effort solution for dynamic environments (Tang et al., 2025).

Despite recent progress, achieving reliable 3D localization in unstructured field environments remains hindered by key challenges. Existing perception methods are sensitive to occlusion, multi-camera calibration under wide baselines or limited overlap faces a trade-off between accuracy and efficiency, and hand-eye calibration often lacks robustness and deployability. These issues limit the effective transformation of visual data into precise robotic actions.

To address these limitations, this study developed a multi-view 3D localization system tailored for agricultural scenarios and makes two key contributions:

- (1) A two-stage calibration method (TCC-RM) that combined rigid initialization with residual modeling was proposed to correct nonlinear distortions and significantly improve the calibration accuracy.
- (2) The proposed multi-RGB-D fusion pipeline enhanced strawberry localization accuracy and expanded perception coverage through occlusion-aware matching and optimized coordinate transformations.

2. Methodology

2.1. Overall framework and hardware configuration of the strawberry harvesting system

This study focused on table-top grown strawberries and proposed a multi-view RGB-D vision-based robotic system for this task. The entire system architecture, illustrated in Fig. 1, consisted of three core functional modules—Perception, Processing, and Localization—implemented on the newly developed robotic harvesting platform, HarvestFlex (<https://xiong-lab.cn/>), as shown in Fig. 2. HarvestFlex integrated a mobile base, vision system, robotic arm, and end-effector.

Perception Module: The vision system adopted a dual RGB-D camera setup (Intel RealSense D455F), symmetrically mounted on both sides of the mobile base to create a complementary field-of-view (FOV) configuration, with the robotic arm's motion rail centrally positioned between the cameras. The cameras were optimally oriented at an outward-facing horizontal angle of approximately 110° and an upward tilt of about 55° , effectively covering the robot's operational workspace, typically ranging from 1.2 to 3 m. RGB and depth images were acquired at 640×480 pixels to balance detail and real-time performance.

Processing Module: This module performed dual-camera and hand-eye calibrations to unify heterogeneous data from RGB images, depth maps, and robot poses. We propose a two-stage TCC-RM calibration (Section 2.2), combining checkerboard initialization with residual modeling for high accuracy. This is followed by a dedicated coordinate transformation study (Section 2.3) that implements and compares CF and BF strategies.

Localization Module: The 3D bounding box of each strawberry was generated from 2D detections and depth information. A modified IoU metric refined cross-view matching, while a weighted multi-view fusion integrated complementary observations (Section 2.4). The final 3D coordinates were transmitted to the robotic system for precise manipulation and harvesting.

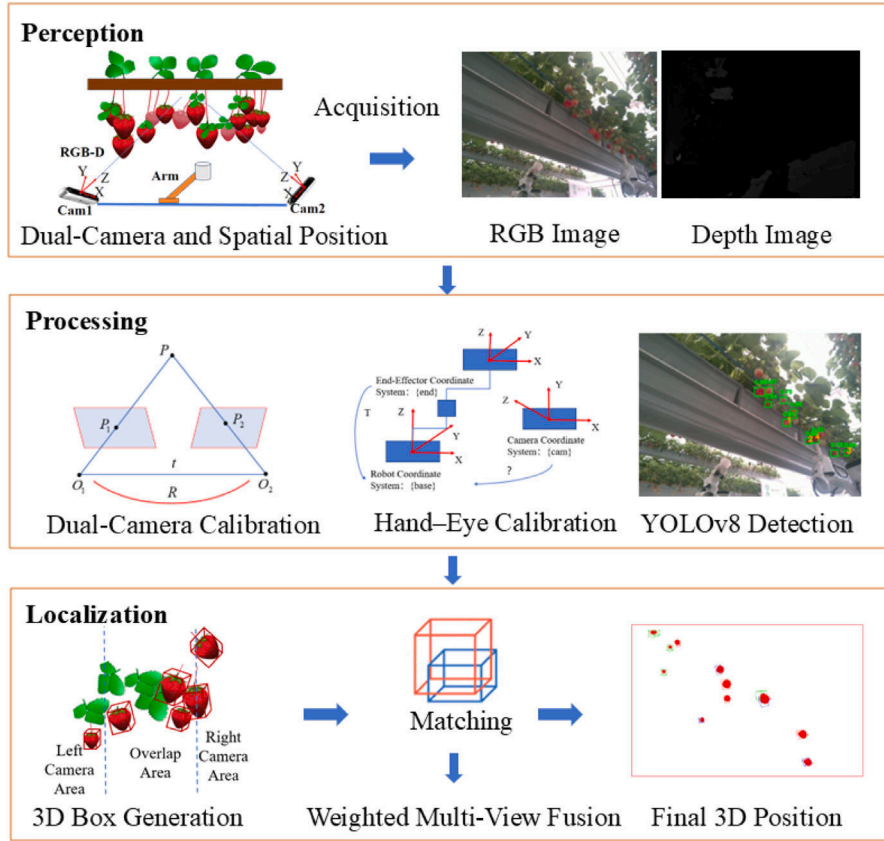


Fig. 1. System architecture of the proposed harvesting framework.

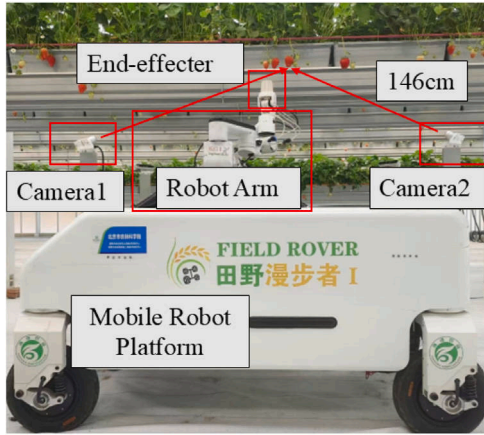


Fig. 2. HarvestFlex robotic harvesting platform (<https://xiong-lab.cn/>).

2.2. TCC-RM: A two-stage calibration method

Camera calibration is essential for geometric consistency across sensors and the robotic arm. While traditional methods like Zhang's and Tsai-Lenz provide global alignment, they fail to correct nonlinear errors, especially prominent in RGB-D cameras due to depth-dependent biases and edge distortions. Pure error compensation methods, though capable of modeling such biases, often lack generalizability without structural constraints.

To address these issues, we proposed a unified two-stage calibration framework, two-sided checkerboard calibration with residual modeling (TCC-RM) (Fig. 3). The first stage established rigid transformations for global consistency, while the second stage employed residual modeling

to explicitly compensate for depth-related errors. This hybrid approach combined structural stability with precise bias correction, providing a millimeter-accurate foundation for multi-view fusion, as validated in Section 3.

2.2.1. Stage 1: Rigid initialization

The first stage of the TCC-RM framework aims to establish the initial extrinsic parameters. Two types of transformations are involved:

Dual-camera calibration employed a novel two-sided checkerboard design, enabling simultaneous feature capture from both cameras. This approach minimized occlusion and improved data collection efficiency. Using Zhang's method (Zhang, 2000), the relative pose between the cameras was computed by leveraging the known rigid transformation between the checkerboard sides. To minimize calibration drift, multiple checkerboard poses covering diverse spatial configurations were sampled, and the pose with the minimum average re-projection error was selected to determine the final transformation matrix T . This matrix establishes the coordinate mapping foundation for subsequent sphere-based error modeling.

Hand-eye calibration adopted an *eye-to-hand* configuration. Using the Tsai-Lenz method (Tsai et al., 1989), the $AX=XB$ problem was solved by combining robot end-effector poses with camera observations of the calibration target, yielding the initial camera-to-robot transformation.

While this stage provided a consistent geometric foundation, it remained susceptible to sensor noise, depth-dependent biases, and nonlinear distortions—limitations that were addressed in the subsequent residual modeling stage.

2.2.2. Stage 2: Sphere-based residual compensation modeling

(a) **Sphere-Based Error Data Acquisition.** To quantify the calibration errors in the dual-RGB-D system, a 3D-printed sphere with a

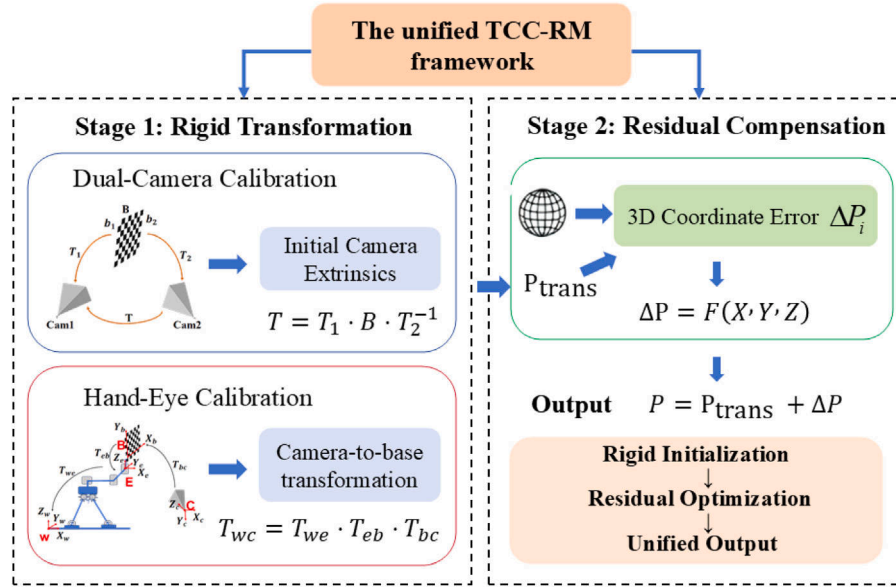


Fig. 3. Flowchart of the TCC-RM calibration method.

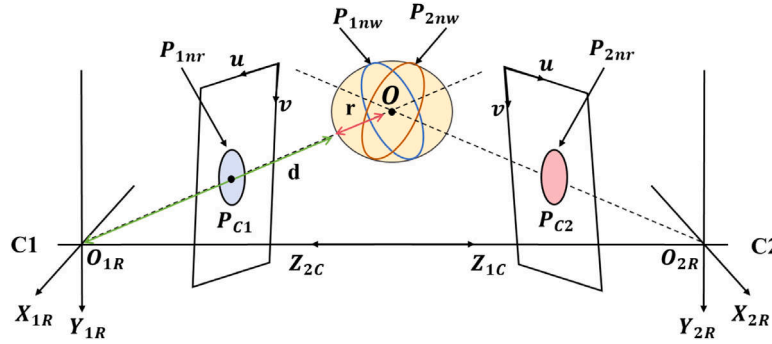


Fig. 4. The figure illustrates the imaging model of cameras C1 and C2 observing a sphere with center O and radius r . Image projections P_{C1} and P_{C2} , along with depth d , allow back-projection to 3D surface points. Adding r recovers the sphere center. $P_{1nr}, P_{1nw}, P_{2nr}, P_{2nw}$ indicate boundary projections; X_{1R}, Y_{1R}, Z_{1C} and X_{2R}, Y_{2R}, Z_{2C} represent the camera coordinate axes. Arrows indicate the positive directions of each axis, following the right-handed coordinate convention.

radius of 5 cm was employed as the calibration target. The method is based on the geometric invariance of the sphere center in multi-view imaging: after rigid transformation, the estimated sphere centers from the two camera coordinate systems should ideally coincide, and the actual measured spatial deviation serves as a direct observation of the calibration error (Fig. 4).

High-precision sphere center extraction was critical for obtaining reliable observational data. As illustrated in Fig. 5, a processing pipeline fusing color and depth information was designed. The procedure began with image pre-processing, where RGB images were rectified using the camera intrinsic matrix and distortion coefficients, and regions of interest were cropped. The calibration sphere was then segmented in the HSV color space using thresholding to extract its complete contour. A minimum enclosing circle was fitted to the contour to estimate the 2D projection center and pixel radius. For 3D reconstruction, the corresponding depth value was retrieved from the aligned depth map, and the 3D coordinates of the surface point were calculated via the pinhole model. Perspective distortion was subsequently corrected using the known sphere radius, yielding the final accurate 3D sphere center coordinates.

This pipeline, through multi-modal data fusion and geometric constraints, ensured the provision of highly accurate and robust sphere center coordinate data for subsequent residual compensation.

(b) Residual Compensation. To address depth-dependent biases and nonlinear distortions not corrected by rigid calibration, a residual

compensation model was introduced. This step refined the coordinate mapping point-wise while preserving global consistency.

For a sphere center observed in frames A and B, the residual error was defined as:

$$\Delta \mathbf{P}_i = \mathbf{P}_i^{(A)} - T_{B \rightarrow A} \mathbf{P}_i^{(B)} \rightarrow \mathbf{A}, \mathbf{P}_i^{(B)}. \quad (1)$$

Each component of $\Delta \mathbf{P}_i$ was approximated via second-order polynomial regression:

$$\Delta x = f_x(x, y, z), \quad \Delta y = f_y(x, y, z), \quad \Delta z = f_z(x, y, z). \quad (2)$$

For any new point $\mathbf{X}^{(B)}$, the compensated coordinate in frame A was calculated as:

$$\hat{\mathbf{X}}^{(A)} = T_{B \rightarrow A} \mathbf{X}^{(B)} + \Delta \mathbf{P}. \quad (3)$$

This formulation applied uniformly to both inter-camera and hand-eye calibration, significantly improving spatial accuracy to millimeter level.

2.3. Coordinate transformation sequence study of the multi-RGB-D-Arm

Existing studies have primarily addressed multi-camera calibration and hand-eye calibration separately (Allegro et al., 2024; Kroeger et al., 2019). For integrated systems, however, the optimal data fusion sequence remains underexplored. To bridge this gap, we analyzed

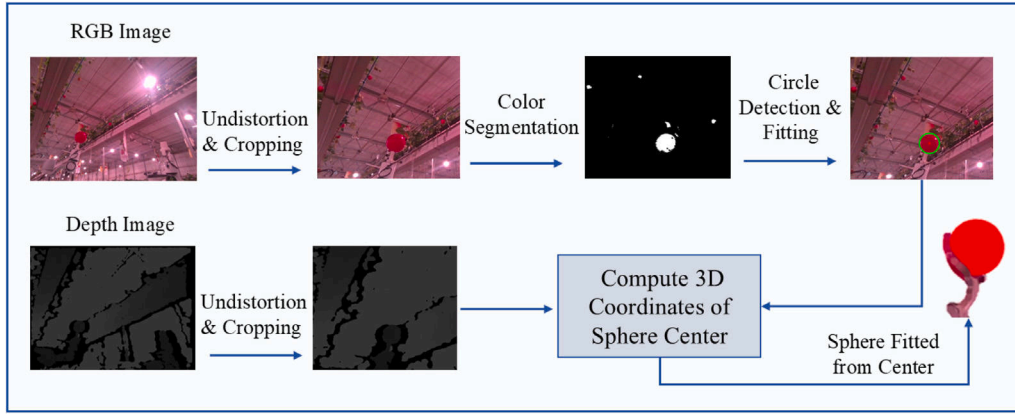


Fig. 5. Sphere center extraction process.

Camera-Frame Fusion(CF):

$$\text{Step 1: } \mathbf{P}_{c_1}^{c_2} = \mathbf{T}_{c_2}^{c_1} \mathbf{P}_{c_2} + \Delta_{c_1}^{(2)}(\mathbf{T}_{c_2}^{c_1} \mathbf{P}_{c_2}, \mathbf{e}) \quad // \text{ Transform RGB-D 2 point to RGB-D 1 with compensation} \quad (4)$$

$$\text{Step 2: } \mathbf{P}_{c_1}^{\text{fused}} = \mathcal{F}(\mathbf{P}_{c_1}, \mathbf{P}_{c_1}^{c_2}) \quad // \text{ Fusion in RGB-D 1 coordinate system} \quad (5)$$

$$\text{Step 3: } \mathbf{P}_b^{\text{rigid}} = \mathbf{T}_{c_1}^b \mathbf{P}_{c_1}^{\text{fused}} \quad // \text{ Rigid transform to robot base frame} \quad (6)$$

$$\text{Step 4: } \mathbf{P}_A = \mathbf{P}_b^{\text{rigid}} + \Delta_b^{(2)}(\mathbf{P}_b^{\text{rigid}}, \mathbf{e}) \quad // \text{ Final base frame compensation} \quad (7)$$

Base-Frame Fusion(BF):

$$\text{Step 1a: } \mathbf{P}_b^{c_1} = \mathbf{T}_{c_1}^b \mathbf{P}_{c_1} + \Delta_b^{(2)}(\mathbf{T}_{c_1}^b \mathbf{P}_{c_1}, \mathbf{e}) \quad // \text{ Rigid + residual compensation for RGB-D 1} \quad (8)$$

$$\text{Step 1b: } \mathbf{P}_b^{c_2} = \mathbf{T}_{c_2}^b \mathbf{P}_{c_2} + \Delta_b^{(2)}(\mathbf{T}_{c_2}^b \mathbf{P}_{c_2}, \mathbf{e}) \quad // \text{ Rigid + residual compensation for RGB-D 2} \quad (9)$$

$$\text{Step 2: } \mathbf{P}_B = \mathcal{F}(\mathbf{P}_b^{c_1}, \mathbf{P}_b^{c_2}) \quad // \text{ Fusion in base frame} \quad (10)$$

Box 1.

and compared two representative transformation strategies—Camera-Frame Fusion (CF) and Base-Frame Fusion (BF)—focusing on how the sequence of coordinate transformations impacts multi-view data fusion. CF first constructs a unified camera coordinate system before transformation to the robot base, whereas BF projects each camera's data directly to the base frame prior to fusion. The mathematical formulations are as follows (See Box 1):

where:

- $\mathbf{P}_{c_1}, \mathbf{P}_{c_2}$: Points detected in RGB-D 1 and RGB-D 2 coordinate systems, respectively.
- $\mathbf{P}_{c_1}^{c_2}$: Point from RGB-D 2 transformed into the coordinate system of RGB-D 1.
- $\mathbf{T}_{c_1}^b, \mathbf{T}_{c_2}^b$: Rigid transformations from RGB-D 1 and RGB-D 2 to robot arm base.
- $\mathbf{T}_{c_2}^{c_1}$: Rigid transformation from RGB-D 2 to RGB-D 1.
- $\Delta_{c_1}^{(2)}(\cdot, \mathbf{e}), \Delta_b^{(2)}(\cdot, \mathbf{e})$: Second-order residual compensation terms in RGB-D 1 or base frame.

As summarized in Table 1, CF provides superior projection consistency through hierarchical compensation but is sensitive to calibration errors. BF offers better modularity and handles view asymmetry well, but requires precise hand-eye calibration. Therefore, in practical deployment, the choice depends on application-specific needs for accuracy versus robustness.

2.4. Strawberry matching and multi-view fusion

To integrate detections from dual RGB-D cameras, we developed a multi-view fusion strategy comprising detection alignment, matching, and weighted fusion. For enhanced robustness in occluded agricultural environments, we introduced: (i) a modified 3D IoU metric tolerant to partial occlusion, and (ii) a weighted fusion scheme incorporating projection area and depth reliability. These adaptations were essential for achieving stable and accurate multi-view localization in practice.

2.4.1. Strawberry matching

In the dual-RGB-D configuration, the same strawberry may be observed from multiple viewpoints, resulting in multiple 3D boxes that provide complementary information of the same target. To integrate these observations, a cross-view matching process is applied, with candidate pairs filtered based on a center-distance threshold. For the remaining pairs, spatial overlap is evaluated using the modified 3D Intersection-over-Union (IoU), which is widely used as an evaluation metric in object detection (Yu et al., 2019), but here adapted as a geometric criterion for cross-view matching.

The traditional 3D IoU often produces low similarity scores under viewpoint changes or partial occlusion. To address this issue, a modified 3D IoU metric is used, which is more tolerant to occlusion and size discrepancies. The geometric relationship is illustrated in Fig. 6, where the intersection volume V_{inter} between two 3D bounding boxes is highlighted. The modified IoU is defined as:

$$\text{IoU}_{\text{new}} = \frac{V_{\text{inter}}}{\min(V_1, V_2)}. \quad (11)$$

Table 1
Integrated comparison of Camera-Frame Fusion (CF) and Base-Frame Fusion (BF) strategies.

Dimension	Camera-Frame Fusion (CF)	Base-Frame Fusion (BF)
Geometric Consistency	Fusion in unified camera coordinate: $P_{\text{fused}} = F(P_{c_1}, T_{c_1}^{c_2} P_{c_2})$ - Accurate XY-plane projection - Depends on inter-camera calibration - Sensitive to depth	Fusion in robot base frame: $P_B = F(T_{c_1}^b P_{c_1}, T_{c_2}^b P_{c_2})$ - Good spatial consistency - Robust to asymmetric views - Inconsistency with poor hand-eye calibration
Compensation Structure	Hierarchical: - $\Delta^{(2)}$: camera frame residual - $\Delta_{b_r}^{(2)}$: base frame residual - Requires joint optimization	Distributed: - Per-branch residual terms - Independent sensor-level correction - May fragment under view inconsistency
Error Controllability	Serial propagation: $P_{c_2} \rightarrow T_{c_2}^{c_1} P \rightarrow P_{\text{fused}} \rightarrow T_{c_1}^b P$ - $\epsilon_{\text{total}} = \epsilon_{\text{cross-cal}} + \epsilon_{\text{fusion}} + \epsilon_{\text{eye-hand}}$ - High single-point failure risk	Parallel propagation: $\epsilon_k = f(\epsilon_{\text{eye-hand}}, \epsilon_{\text{sensor}})$ - Modular, branch-specific errors - Scattered or inconsistent residual paths
Mathematical Stability	- Requires planar convexity for fusion - Coupled Jacobian-based modeling - Sensitive to depth discontinuity	- Uses least-squares in Euclidean space - Allows independent optimization - Only needs geometric consistency - May affect convergence if residuals mismatch

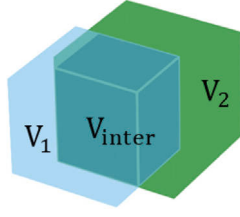


Fig. 6. Schematic diagram of strawberry 3D box intersection.

Where V_1 and V_2 represent the volumes of the two bounding boxes. This approach ensures that occlusion and partial visibility do not disrupt the matching process.

2.4.2. Multi-view fusion strategy

To enhance positional accuracy in multi-view fusion, both the target's projection area in RGB images and its depth information were incorporated into the weight design. A larger projection area indicates more reliable localization under the given viewpoint, while a smaller depth value reflects higher measurement precision. These attributes jointly represent observation quality.

Formally, given center points c_1, c_2 , projection areas A_1, A_2 , and depths d_1, d_2 , the fusion weights are defined as:

$$w_i = \frac{A_i}{d_i^2 + \epsilon}, \quad (i = 1, 2), \quad (12)$$

where $\epsilon > 0$ ensures numerical stability. The fused center point is then obtained as:

$$c_{\text{fused}} = \frac{w_1 c_1 + w_2 c_2}{w_1 + w_2}. \quad (13)$$

When both weights approach zero due to very small projection areas or unstable depth estimates, the fused center was instead approximated by the simple average of c_1 and c_2 to avoid abnormal predictions.

2.4.3. Hybrid localization strategy for computational efficiency

The overall process, from recognition to fusion, is illustrated in Fig. 7. To maximize computational efficiency, a hybrid localization strategy was implemented based on the fixed geometric configuration of the dual cameras. Prior to operation, the non-overlapping regions of the two cameras' fields of view were calculated. During runtime, strawberries detected within these pre-defined non-overlapping regions were directly localized using single-camera data, effectively bypassing computationally expensive cross-view matching. For detections in the

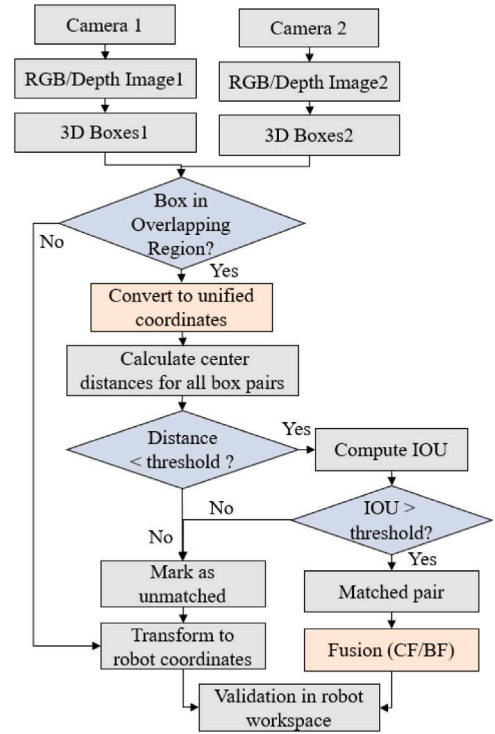


Fig. 7. Strawberry Fusion Process.

overlapping region, a modified 3D Intersection-over-Union (IoU) metric was used for matching. Successfully matched pairs were then processed with multi-view fusion to achieve superior localization accuracy. All resulting 3D positions—whether from efficient single-camera localization or accurate multi-view fusion—were transformed into the robot-base frame for picking. This strategy ensures optimal trade-off between computational speed and localization precision.

3. Experiments and results

Experiments were conducted on the HarvestFlex strawberry-harvesting robot. The evaluation focused on three main aspects: the calibration accuracy of the proposed TCC-RM method for dual-RGB-D and hand-eye systems; the robustness of the improved IoU-based matching strategy in multi-fruit scenarios; and the 3D localization



Fig. 8. Checkerboard and calibration sphere images in dual-RGB-D calibration. (a) and (b) show checkerboard views from Camera 1 and 2 for initial extrinsic calibration; (c) and (d) show calibration sphere views for center extraction and residual error compensation.

Table 2

Error reduction achieved by the TCC-RM method in dual-RGB-D calibration. “Euc.” denotes the Euclidean error computed from the XYZ axes.

Method	MAE (mm)				STD (mm)				MAX (mm)				RMSE (Euc., mm)
	X	Y	Z	Euc.	X	Y	Z	Euc.	X	Y	Z	Euc.	
TCC	8.64	7.97	5.79	13.68	3.73	2.84	3.28	4.06	16.52	12.29	11.41	21.11	14.24
TCC-RM	4.37	2.49	3.41	6.70	2.93	1.65	2.19	2.79	9.69	5.86	8.49	13.15	7.23
Reduction (%)	49.42	68.76	41.11	51.02	21.45	41.90	33.23	31.28	41.34	52.32	25.59	37.71	49.23

performance under both unoccluded and partially occluded conditions. Additionally, perception coverage was specifically evaluated in multi-strawberry scenes to assess the system’s effectiveness in expanding visible target regions.

3.1. Calibration performance evaluation

To validate the effectiveness of the proposed TCC-RM calibration method, we conducted experiments for both dual-RGB-D camera extrinsic calibration and camera-to-robot-base (hand-eye) calibration. Fig. 8 showed images of the checkerboard and calibration sphere from the two camera views. The effectiveness of the method was systematically evaluated at three levels: error statistics, sample trends, and error distribution. Quantitative analysis was based on four standard error metrics: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Standard Deviation (STD), and Maximum Error (MAX), where MAE and RMSE reflect overall error magnitude, while STD and MAX capture error dispersion and outlier behavior (Hodson, 2022). These metrics were applied to the Euclidean error in the estimated 3D position of a target point (e.g., calibration sphere center), as well as to its decomposition along the X, Y, and Z axes.

3.1.1. Dual-RGB-D calibration accuracy

For the dual-RGB-D calibration task reported in Table 2, the calibration error refers to the Euclidean distance between the sphere center position directly observed in the reference camera’s coordinate frame and the position transformed from the other camera, as well as its components along the X, Y, and Z axes. As quantified in Table 2, TCC-RM achieves substantial absolute reductions in all error metrics. Most critically, the Euclidean RMSE was reduced by 7.01 mm (from 14.24 mm to 7.23 mm, a 49.2% reduction). A major improvement is seen in the Y-axis MAE, which saw an absolute decrease of 5.48 mm (from 7.97 mm to 2.49 mm). Significant absolute error reductions were also achieved on the X-axis (4.27 mm) and Z-axis (2.38 mm). These improvements, particularly on the Y-axis, are potentially linked to the effective compensation of systematic biases related to sensor placement and the camera’s field-of-view limitations.

Fig. 9 illustrates per-sample error trends. Post-compensation curves (red) exhibit enhanced stability across all axes with fewer outliers. Y-axis and Euclidean errors cluster tightly near zero, indicating effective suppression of systematic bias. Fig. 10 provides statistical validation: Kernel Density Estimation (KDE) curves shift toward zero and narrow significantly after compensation. The Euclidean error mean decreased

from 13.68 mm to 6.70 mm (51.0% reduction), with standard deviation dropping from 4.06 mm to 2.79 mm (31.28% reduction). Y-axis improvements dominate, aligning with its MAE reduction.

Collectively, Table 2, Figs. 9, and 10 demonstrate TCC-RM’s efficacy in modeling directionally structured residuals—particularly for Y-axis correction. This validates its physical and geometric rationality while highlighting robustness in multi-camera calibration scenarios.

3.1.2. Hand-eye calibration accuracy

TCC-RM delivered even more pronounced improvements in hand-eye calibration between Camera 1 and the robotic arm (Table 3, Fig. 11), reducing errors to the millimeter level. For the hand-eye calibration task reported in Table 3, the calibration error refers to the Euclidean distance between the sphere center’s position transformed from the camera to the robot base frame and its ground-truth position in the robot base frame, as well as the errors along the X, Y, and Z axes. The most dramatic absolute improvement was on the X-axis, where the MAE was cut by 16.51 mm (from 18.58 mm to 2.07 mm) and the standard deviation was reduced by 3.29 mm. This optimization stems from the effective modeling and compensation of major systematic biases, such as initial sensor misalignment. The Z-axis MAE was also substantially reduced by 5.54 mm (65.1% reduction), effectively suppressing depth residuals caused by quantization noise and geometric distortion (Fig. 11). On the Y-axis, where the baseline MAE was already relatively low (5.43 mm), the absolute improvement was smaller (0.48 mm), yet the method still improved stability, as evidenced by a reduction in standard deviation (1.20 mm). It is noteworthy that the error for the TCC-RM method on the Y-axis is larger than that of the TCC method in some samples (Fig. 11). This is likely because the data used to build the compensation model had a relatively limited range of variation along the Y-axis, while the validation data exhibited greater variation in this axis. Nevertheless, the TCC-RM method demonstrates significantly higher overall robustness.

KDE-based error distributions (Fig. 12) further confirm: post-compensation errors shifted closer to zero with tighter concentration. Euclidean error mean decreased from 21.42 mm to 6.50 mm, and STD dropped from 4.78 mm to 1.84 mm. The most pronounced KDE curve narrowing occurred for X-axis and Euclidean errors, demonstrating TCC-RM’s capability to compensate structured residuals and improve calibration robustness.

TCC-RM achieved Euclidean RMSEs of 7.23 mm (dual-RGB-D) and 6.74 mm (hand-eye), meeting millimeter-level accuracy requirements for fruit localization. These results validate residual modeling as a practical and scalable error compensation strategy for agricultural vision systems.

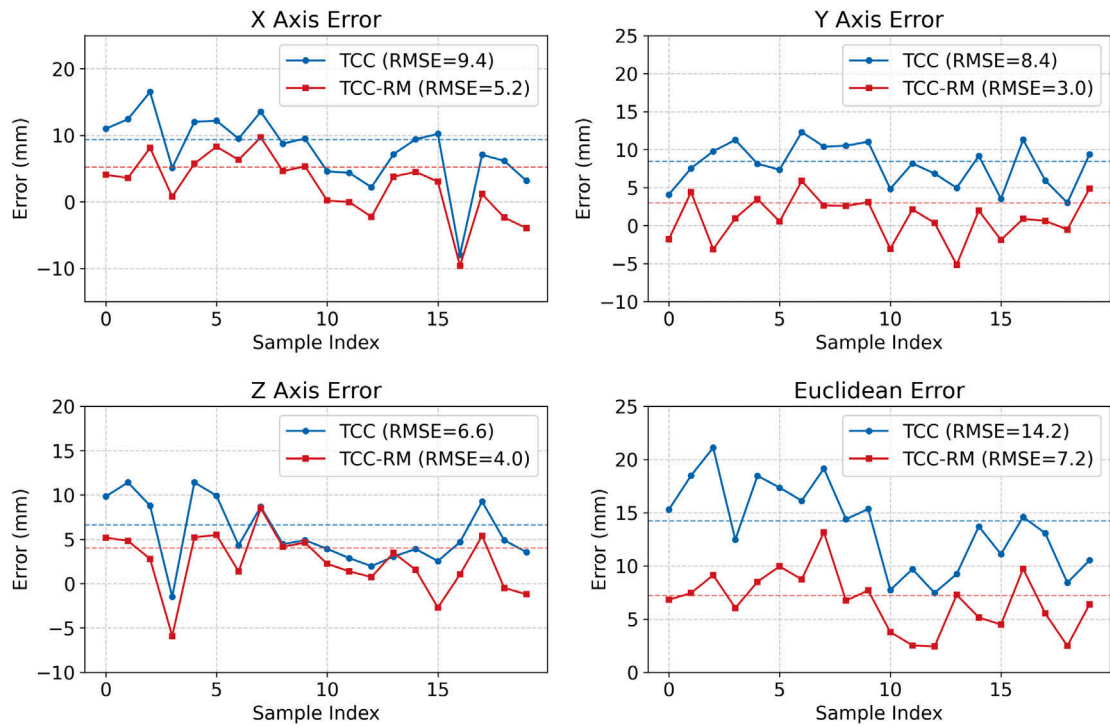


Fig. 9. Per-sample error trends in dual-RGB-D calibration using TCC-RM. Each subplot shows error changes on the X, Y, Z axes and Euclidean distance. Blue: before compensation (TCC); red: after compensation (TCC-RM).

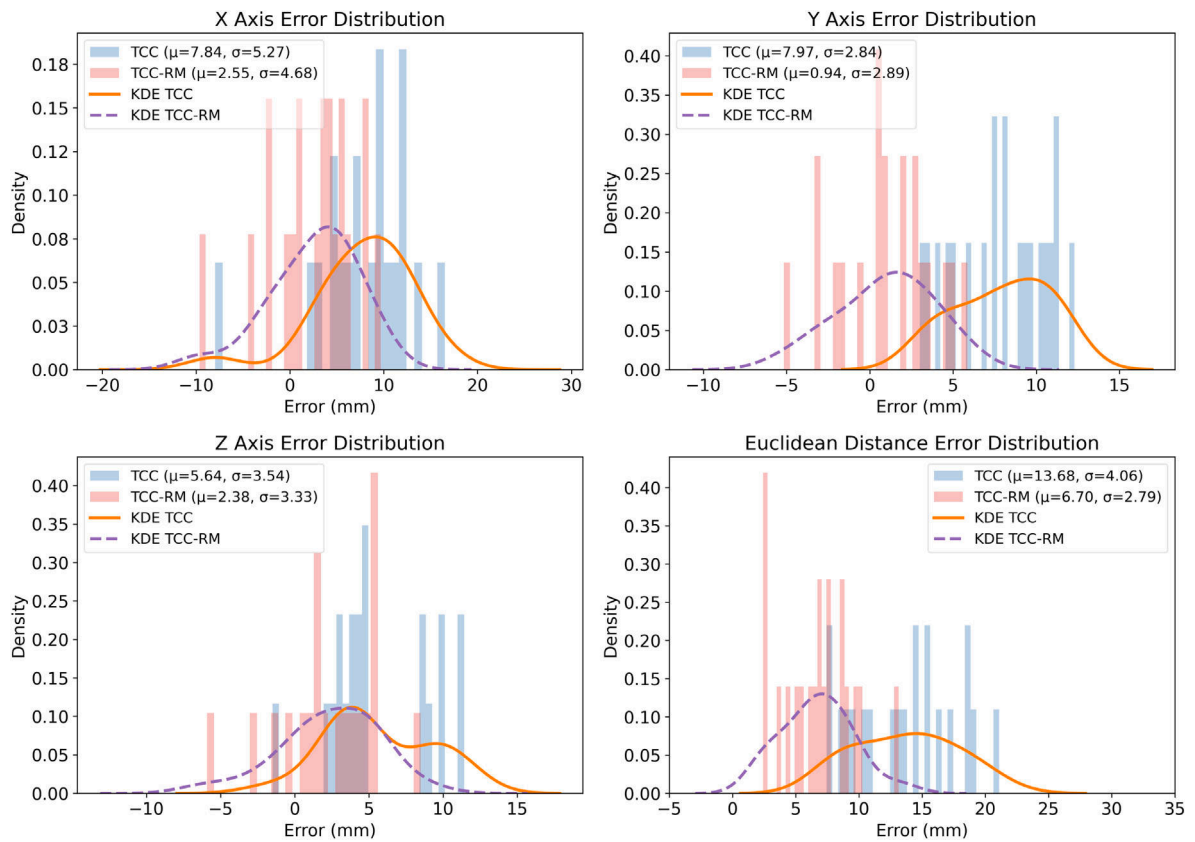


Fig. 10. KDE-based comparison of dual-RGB-D calibration errors on the X, Y, Z axes and Euclidean distance. Orange and purple curves represent errors before and after compensation, respectively. μ and σ denote the mean and standard deviation.

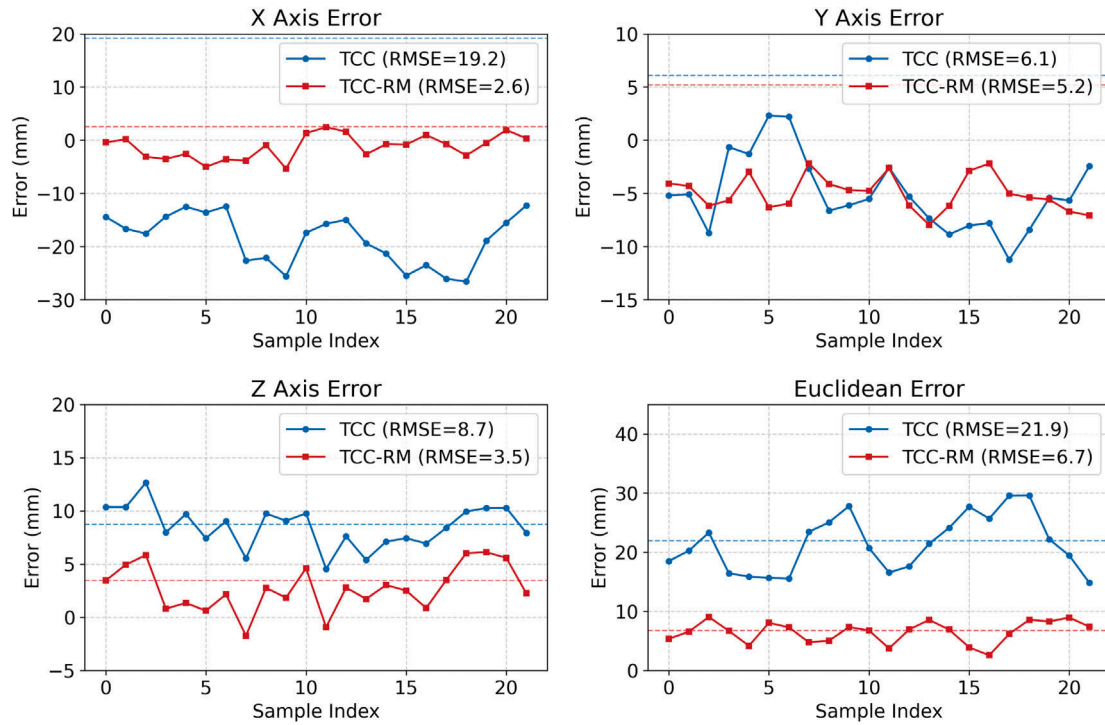


Fig. 11. Per-sample error trends for TCC-RM in hand-eye calibration. Blue: before compensation (TCC); red: after compensation (TCC-RM), across X, Y, Z axes and Euclidean distance.

Table 3

Error reduction achieved by the TCC-RM method in hand-eye calibration.

Method	MAE (mm)				STD (mm)				MAX (mm)				RMSE (Euc.mm)
	X	Y	Z	Euc.	X	Y	Z	Euc.	X	Y	Z	Euc.	
TCC	18.58	5.43	8.51	21.42	4.82	2.83	1.95	4.78	26.57	11.22	12.65	29.59	21.92
TCC-RM	2.07	4.95	2.97	6.50	1.53	1.63	1.81	1.84	5.34	7.95	6.11	9.04	6.74
Reduction (%)	88.86	8.84	65.10	69.64	68.27	42.40	7.18	61.51	79.89	29.15	51.70	69.45	69.25

Table 4

Average performance results of different IoU matching, IoU threshold is 0.4 for both.

Matching strategy	Precision (%)	Recall (%)	F1-score (%)
Traditional IoU	100.00	19.75	32.99
Modified IoU	93.65	72.84	81.94

3.2. Performance of modified IoU matching under occlusion

Robust matching across viewpoints under occlusion is critical for multi-view fusion. We evaluated the proposed modified IoU strategy against the traditional IoU in 21 multi-view strawberry samples exhibiting mild occlusion. The IoU threshold was set to 0.4 for both strategies. The performance difference between the two strategies is systematically evaluated from three dimensions: Precision, Recall, and F1 score.

Table 4 summarizes the average performance comparison results of the two strategies. Traditional IoU has high precision (100%) but very low recall (19.75%), indicating that although this method has almost no false positives, it misses a large number of real strawberries, and the F1 score is low (32.99%), reflecting poor overall performance. Modified IoU has high precision (93.65%), recall significantly improved (72.84%), and a higher F1 score (81.94%), indicating that this method maintains a low false positive rate while reducing missed detections, with overall performance superior to traditional IoU.

Fig. 13 illustrates a sample-by-sample comparison of matched strawberries across the 21 test sets. The traditional IoU strategy could only

match 1–2 targets in most samples, with significant missed detections. In contrast, the improved strategy matched over 4 strawberries in several samples, closer to the true count, demonstrating stronger matching ability. In typical scenes like samples 4, 5, and 7, the improved strategy perfectly matched all true targets, showing higher accuracy and robustness.

The modified IoU strategy, leveraging minimum volume normalization, substantially enhanced the robustness of instance association under occlusion, reducing missed matches by 53.1% (Recall increase) and improving overall matching quality (F1-score increased 48.95%). This is vital for reliable multi-view data fusion.

3.3. Strawberry localization accuracy evaluation (Occlusion-free & occluded)

The core objective of the perception system is accurate 3D localization. We evaluated the performance of the proposed Camera-Frame Fusion (CF) and Base-Frame Fusion (BF) strategies against single-camera baselines in both occlusion-free and occluded multi-strawberry scenarios.

All experimental data were collected in real-world agricultural fields under varying lighting conditions, representative of typical agricultural scenarios. The strawberry localization error is the difference in 3D coordinates under the base frame between the ground-truth position (obtained through forward kinematics) and the vision-based estimate, specifically referring to the coordinates of the strawberry's center point. To obtain the true position of the strawberries, a 3D-printed fixture was

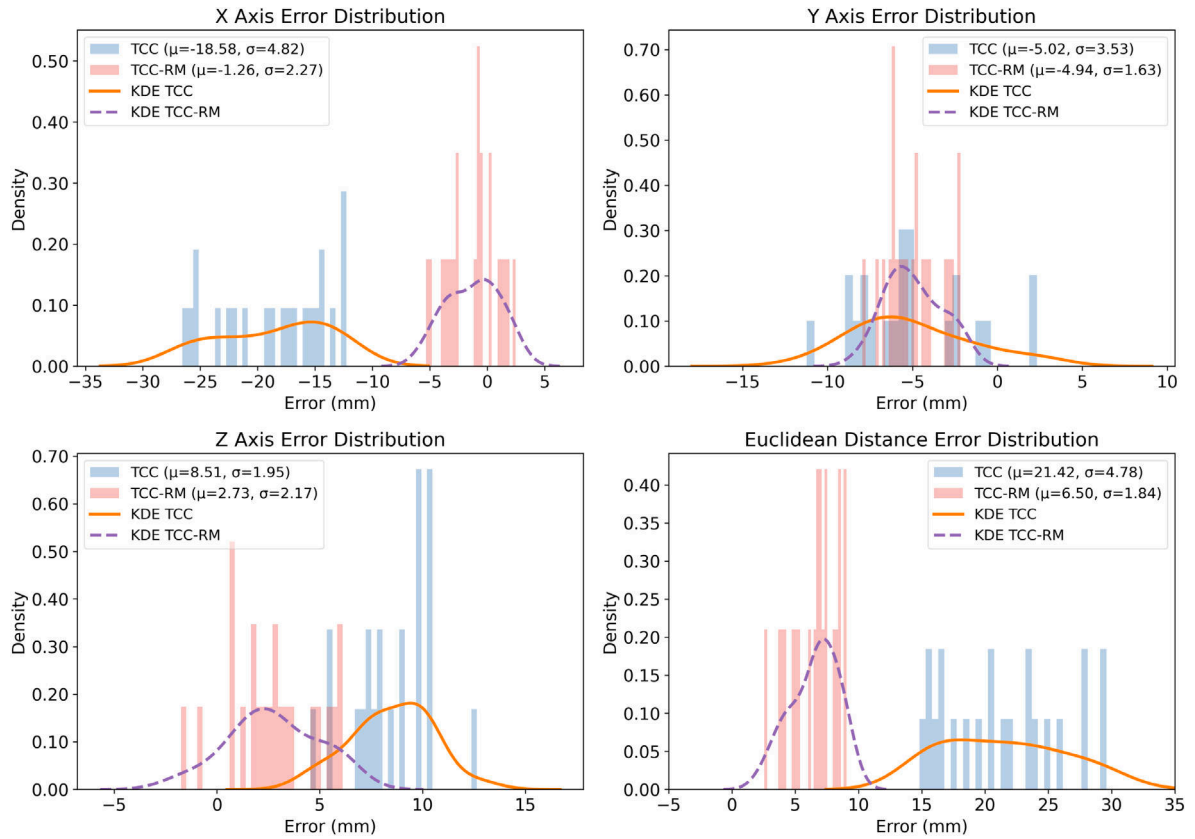


Fig. 12. KDE-based distribution of hand-eye calibration errors on the X, Y, Z axes and Euclidean distance. Orange: before compensation (TCC); purple: after (TCC-RM). μ and σ denote the mean and standard deviation.

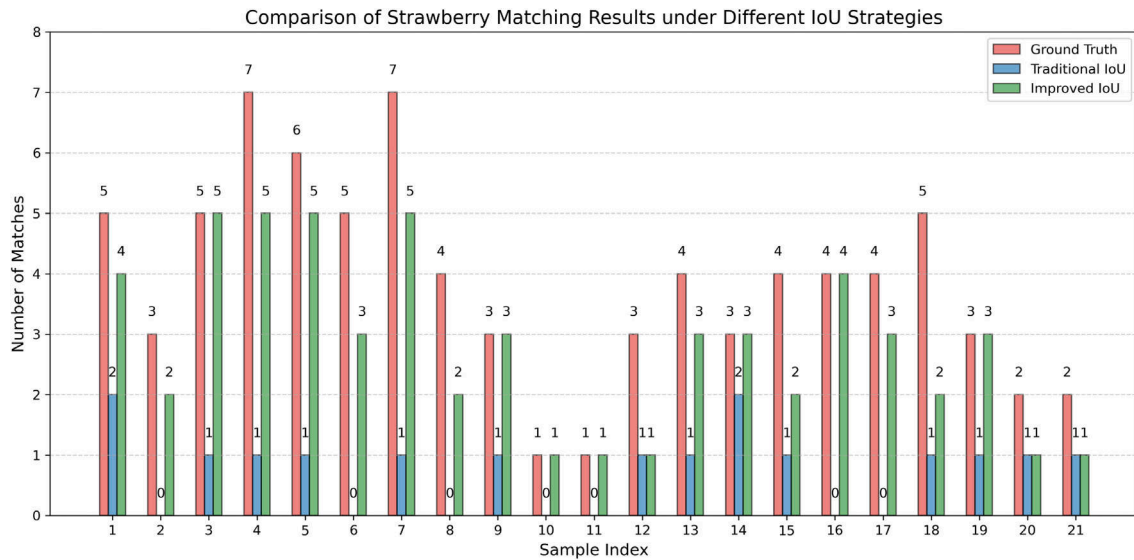


Fig. 13. Comparison of strawberry sample matching under different IoU strategies. Red bars indicate ground truth, blue bars represent traditional IoU matches, and green bars show results from the improved method.

attached to the robot's end-effector, as shown in Fig. 14. The fixture featured a slender metal pin at its top, which was used to point to the strawberry. The position of the pin's tip could be accurately determined via the robot's inverse kinematics. To simplify the experiment, the various strawberries (under unoccluded and slightly occluded conditions) were directly affixed to the metal pin. The robot arm was then moved to different positions to quantify the localization error.

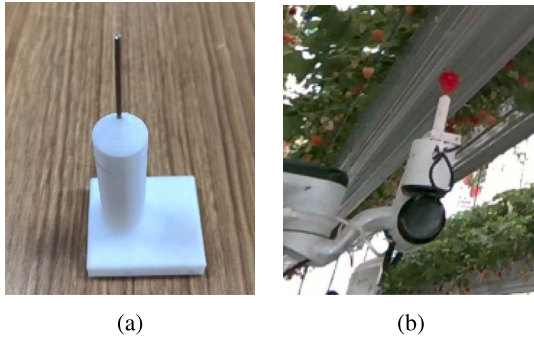
The 3D points of strawberries in the robot's base frame were obtained using five distinct strategies for comparison: the two proposed fusion strategies—CF and BF—alongside three single-camera baseline methods:

- C1-B: Camera 1 to the robotic arm base transformation
- C2-B: Camera 2 to the robotic arm base transformation
- C2-C1-B: Camera 2 to Camera 1 to base

Table 5

Comparison of 3D localization errors under different fusion strategies in unoccluded and occluded scenes.

Scene	Method	MAE (mm)				STD (mm)				MAX (mm)				RMSE (Euc.)
		X	Y	Z	Euc.	X	Y	Z	Euc.	X	Y	Z	Euc.	
Unoccluded	C1-B	3.44	2.12	3.47	6.03	2.99	1.30	1.74	2.38	11.48	5.93	6.43	12.68	6.49
	C2-B	4.37	4.33	3.64	6.26	2.65	2.44	2.05	2.36	12.60	9.47	8.64	14.73	8.52
	C2-C1-B	3.80	4.31	5.45	9.28	2.99	3.35	3.73	2.59	10.90	9.12	12.49	13.68	9.60
	CF	2.48	1.74	4.17	5.83	1.76	1.10	2.55	1.83	6.99	4.17	9.48	10.27	6.11
	BF	2.60	2.17	3.80	5.70	2.03	1.70	1.54	1.66	7.54	5.85	7.50	9.38	5.94
Slight Occluded	C1-B	3.82	1.90	3.85	6.36	2.92	1.42	2.49	3.04	10.54	5.89	9.39	12.54	7.05
	C2-B	5.35	3.52	4.48	8.72	2.91	2.51	2.92	2.88	13.22	11.01	10.73	15.99	9.19
	C2-C1-B	4.73	2.88	7.15	9.98	2.92	2.30	3.47	2.84	13.71	9.08	14.86	17.47	10.38
	CF	2.46	1.57	5.19	6.44	1.97	1.30	2.43	2.33	7.27	4.42	10.86	11.95	6.85
	BF	2.80	2.16	3.75	5.84	2.25	1.73	2.26	2.39	8.02	6.30	9.10	11.15	6.31

**Fig. 14.** Strawberry Positioning Measurement Device.(a) Device schematic.(b) A strawberry in the fixed state.

The efficacy of the proposed fusion strategies is first illustrated qualitatively. Fig. 15 demonstrates successful cross-view matching under slight occlusion: a strawberry partially obscured in Camera 1 is accurately matched in Camera 2 using the proposed fusion. Quantitatively, the results (Table 5) show that both fusion strategies consistently outperform single-camera baselines. For instance, compared to the best single-camera baseline (C1-B), the BF fusion strategy reduced the Euclidean RMSE by 0.55 mm in unoccluded scenes and 0.74 mm under slight occlusion. The percentage reductions across different scenarios and metrics ranged from 2.84% to 39.21%, demonstrating the robustness of the fusion approach. Among all approaches, BF achieves the highest overall 3D accuracy, attaining the lowest Euclidean RMSE (5.94 mm unoccluded, 6.31 mm slight occluded), while CF also performs well (6.11 mm unoccluded, 6.85 mm slight occluded).

The strategies showed complementary strengths: CF excelled in the X-Y plane, reducing horizontal MAE under occlusion by 35.61% (X) and 16.84% (Y) vs. C1-B. BF offered superior depth (Z-axis) stability, achieving the lowest standard deviation (1.54 mm unoccluded; 2.26 mm slight occluded) and MAE (3.75 mm), outperforming CF by up to 27.7%.

Both strategies effectively suppressed extreme errors. BF reduced the maximum Euclidean error by 36.3% (from 14.73 mm to 9.38 mm) in unoccluded scenes and retained a 10.5% RMSE advantage over C1-B under slight occlusion. This improved robustness and consistency under challenging conditions is further illustrated in the occlusion error distributions (Fig. 16). Whereas single-camera baselines exhibit broad, skewed curves with pronounced tails, CF and BF both yield markedly sharper and more concentrated distributions—confirming their enhanced reliability.

Furthermore, we evaluated the localization performance for strawberries under moderate occlusion in a denser field environment. In this scenario, instead of fixing strawberries to the fixture, the tip of the metal pin was positioned directly beneath the center of each occluded

strawberry to establish ground truth, and the localization error was calculated against the visual prediction. Fig. 17 shows the front view of the strawberry scene along with the corresponding images from Camera 1 and Camera 2. Quantitative results (as shown in Table 6) reveal that the BF method achieved the lowest overall RMSE (12.62 mm), demonstrating the best precision in Euclidean localization under occlusion. The CF method followed closely with an RMSE of 13.65 mm, and notably achieved the lowest Euclidean MAE (12.01 mm), indicating high accuracy. In contrast, the C2-C1-B method exhibited the largest errors across almost all metrics, including the highest Euclidean MAE (18.38 mm), MAX error (34.85 mm), and RMSE (25.37 mm), highlighting its limitations in handling occlusions in dense environments. The single-camera methods (C1-B, C2-B) showed intermediate performance, with C1-B yielding a lower Euclidean MAE than C2-B (13.34 mm vs. 13.79 mm), but a comparable RMSE (14.32 mm vs. 14.72 mm).

As occlusion grew from slight to moderate, all methods showed a corresponding increase in localization error. The root cause was two-fold: structured-light cameras provided fewer valid depth points for occluded targets and suffered from point cloud adhesion in dense scenes. Under mild occlusion, enough strawberry pixels remained for a basic 3D approximation. However, moderate occlusion reduced usable data and introduced depth errors that skewed calculations. The incomplete data also impaired filtering, often discarding good points with the noise (Ge et al., 2023). These issues—combined with point cloud adhesion and the flattening of occluded shapes (Zhang, 2018)—confirm the problem lay in the camera's perceptual limits and algorithmic shortcomings, not in calibration. Thus, the resulting 3D bounding boxes were inaccurate, increasing localization error.

Despite this overall performance degradation, the proposed fusion strategies maintained their relative advantage. The BF method achieved the lowest overall RMSE (12.62 mm), while the CF method also demonstrated competitive performance with a lower Euclidean MAE (12.01 mm) than BF (13.00 mm). These results confirm that both fusion frameworks offer superior robustness compared to single-view baselines under challenging occlusion conditions. It should be clarified that our study also attempted to evaluate performance under heavy occlusion; however, the current perception model failed to detect strawberries in such scenarios, preventing a quantitative localization analysis. Therefore, the discussion and results presented here are applicable to conditions ranging from unoccluded to moderately occluded. The reported localization errors represent the cumulative effect of errors from perception, calibration, robotic arm motion, and ground-truth measurement. Meanwhile, the full pipeline—from perception and multi-target matching to base-frame transformation—was executed at an average of 250 ms per frame.

3.4. Multi-view completion and perception range enhancement

In complex agricultural environments, occlusions from fruits, foliage, and limited viewpoints often lead to blind spots in single RGB-D

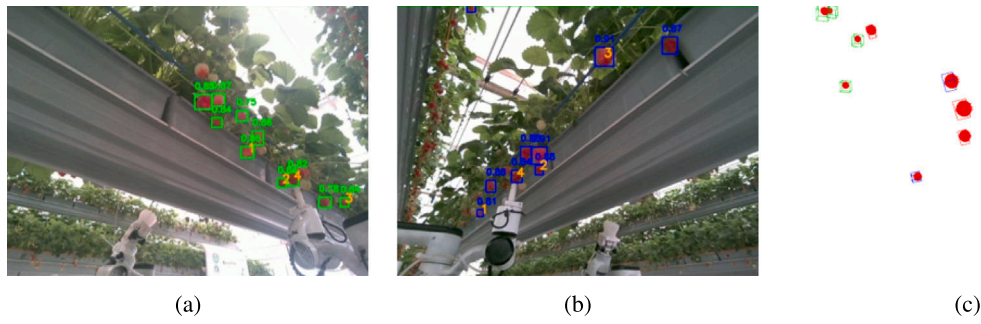


Fig. 15. Matching of a slightly occluded strawberry. (a) Matched strawberry in Cam1 view; (b) Matched strawberry in Cam2 view; (c) 3D matching verification (Green box: Cam1 detection; Blue box: Cam2 detection; Red box: Fusion result).

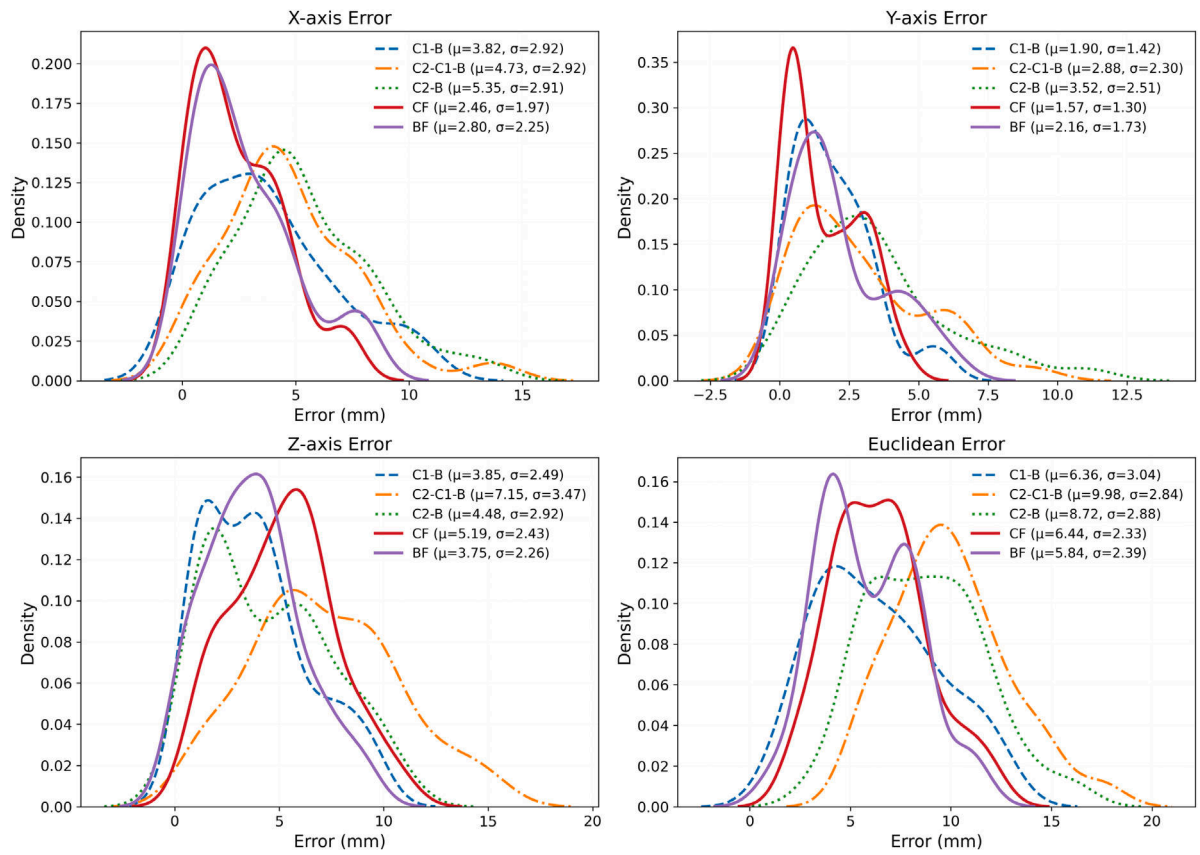


Fig. 16. Probability density distribution of errors under occluded conditions for different localization methods (X, Y, Z, and Euclidean distance). Values in parentheses are mean (μ) and standard deviation σ .

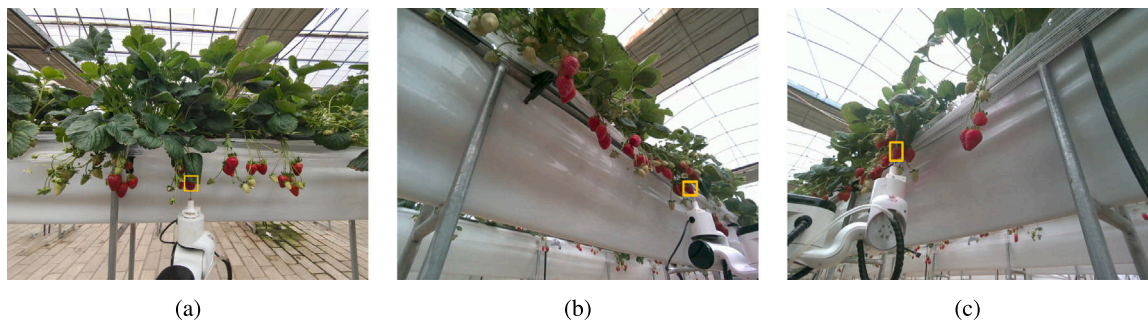


Fig. 17. Strawberry Occlusion under Dense Conditions. (a) Front View; (b) Cam1 view; (c) Cam2 view.

Table 6

Comparison of 3D localization errors of different strawberry localization methods under dense conditions.

Method	MAE (mm)				STD (mm)				MAX (mm)				RMSE (Euc.)
	X	Y	Z	Euc.	X	Y	Z	Euc.	X	Y	Z	Euc.	
C1-B	7.67	5.27	4.36	13.34	5.13	3.62	3.29	4.35	15.79	15.95	13.64	21.31	14.32
C2-B	7.32	9.43	8.26	13.79	5.03	4.50	3.11	5.14	18.08	21.91	10.82	25.87	14.72
C2-C1-B	11.99	9.02	10.66	18.38	7.53	9.13	4.43	9.68	31.31	29.00	16.17	34.85	25.37
CF	5.99	6.19	6.13	12.01	3.38	3.85	2.01	3.86	12.78	18.57	12.01	20.93	13.65
BF	7.08	7.89	5.36	13.00	4.21	4.12	2.23	4.31	21.95	17.45	10.03	20.15	12.62

Table 7

Multi-view perception coverage gain statistics table.

Sample	Strawberry detections (Cam1)	Strawberry detections (Cam2)	Matched pairs	Fused detections	Max single-view detections	Absolute increase	Coverage gain (%)
1	9	9	3	15	9	+6	66.67
2	9	8	2	15	9	+6	66.67
3	12	9	4	16	12	+4	33.33
4	11	6	3	14	11	+3	27.27
5	4	9	2	11	9	+2	22.22
6	6	6	2	10	6	+4	66.67
7	4	4	1	7	4	+3	75.00
8	5	4	1	8	5	+3	60.00
9	7	7	2	12	7	+5	71.43
10	3	6	1	8	6	+2	33.33
11	6	3	1	8	6	+2	33.33
12	9	7	4	12	9	+3	33.33
13	4	9	1	12	9	+3	33.33
14	5	9	2	12	9	+3	33.33
15	7	4	4	12	7	+5	71.43
16	7	9	4	12	9	+3	33.33
17	8	10	2	16	10	+6	60.00
18	10	10	5	15	10	+5	50.00
19	3	10	2	11	10	+1	10.00
20	5	7	2	10	7	+3	42.86
21	6	8	1	13	8	+5	62.50
Total	140	159	49	250	175	+75	42.86

systems, reducing perception completeness and picking success. To address this, we deployed a multi-camera fusion system. To establish a theoretical performance baseline for this system, we first constructed a geometric perception model of the bilateral camera layout (Fig. 2). Under idealized assumptions of continuous target distribution and a perfect detection algorithm, this model calculated an average theoretical perception coverage gain upper bound of approximately 60% across the operational distance range (1.2 m to 3.0 m), representing the maximum potential improvement in physical Field of View (FOV).

However, we recognized that real-world agricultural environments present challenges such as discretely distributed targets and performance degradation at the FOV edges. Therefore, we introduced a task-oriented Perception Coverage Gain metric based on actual detection counts to more accurately reflect the system's effective capability. The evaluation metrics are defined as follows:

- A: Number of strawberries detected by the left camera (Cam1).
- B: Number of strawberries detected by the right camera (Cam2).
- C: Number of matched pairs determined to be the same strawberry by modified IoU matching (threshold 0.5).

Then the non-redundant total detection count of the fused system is: Fused = A + B - C. Using the maximum detection count from a single view as the baseline, define the perception coverage gain rate:

$$\text{Coverage Gain} = \frac{(A + B - C) - \max(A, B)}{\max(A, B)} \times 100\% \quad (14)$$

This metric measured the net increase in total detected targets by the multi-view system, excluding redundancy caused by overlap, and can more objectively reflect the completion capability of the perception

range. No false positives were encountered in this experiment. However, missed detections did occur for smaller strawberries, strawberries under very strong lighting, and those with severe occlusion. Missed detections were not considered during the experiment.

As summarized in Table 7, the multi-view RGB-D fusion system outperformed the single-view baseline, with an average coverage gain of 42.86%. Notably, 85.7% of samples achieved gains above 30%. Each sample yielded an average of 2.33 matched pairs, indicating moderate overlap, while 80.4% of detections came from non-overlapping views, highlighting bilateral complementarity. Some samples (e.g., Sample 5: Cam1=4, Cam2=9) showed large inter-view discrepancies, likely due to asymmetric camera mounting or scene-specific occlusion. Such variations are common in field settings and underscore the benefit of multi-view fusion in mitigating single-view bias.

Overall, the dual-RGB-D setup expands perception range and improves robustness by reducing occlusion-related blind spots.

3.5. Discussions

This study systematically evaluated the performance of the TCC-RM calibration method and the multi-view perception system. Our TCC-RM method demonstrated superior hand-eye calibration accuracy compared to the deep learning-based Kalib method, even when operating under an inherent disadvantage. Using an external configuration, we achieved errors of 2.07/4.95 mm (X/Y), which outperforms Kalib's (Tang et al., 2025) reported errors of 4.8 mm (eye-to-hand) and 3.0 mm (eye-in-hand). This is significant because the external configuration is theoretically less precise than the setups Kalib employs,

making our method's higher observed accuracy a strong testament to its robustness. Furthermore, the illumination sensitivity of the sphere-based residual modeling was addressed by performing this step as a one-time calibration under controlled indoor conditions, ensuring long-term parameter stability during field operation.

In terms of occlusion robustness, the upward-looking side-view camera configuration was employed to circumvent frontal occlusion from leaves and better observe fruits within the canopy. A characteristic of this perspective is the convergence of spatial occlusions (where lower fruits block higher ones) and projection overlaps in the 2D image plane. The multi-view fusion strategy effectively addressed this by integrating geometric information from disparate viewpoints to reconstruct an accurate 3D scene. Consequently, the system significantly enhanced localization stability from unoccluded to moderately occluded conditions. Nonetheless, perception under heavy occlusion remains a fundamental challenge.

The high-precision localization, quantified under static conditions, provides a reliable input for the current open-loop system, where the 250 ms processing time is viable within the 8-second cycle. However, transitioning to dynamic closed-loop harvesting introduces system-level challenges, as final success depends not only on perception accuracy but also on robot control and motion planning. Thus, this work establishes a precise perceptual foundation, while identifying dynamic tracking and extreme occlusion handling as key targets for future development.

4. Conclusion

This study established a high-precision, multi-view RGB-D perception system for strawberry harvesting, delivering a comprehensive solution from calibration to occlusion-aware fusion. The cornerstone of this system was the TCC-RM calibration method, which reduced dual RGB-D and hand-eye calibration errors by 49.23% and 69.25%, respectively. The resulting accuracy enabled a robust fusion pipeline, wherein the Camera-Frame and Base-Frame strategies offered distinct performance trade-offs, both significantly outperforming single-view methods under occluded conditions. The integrated system operates at 4 Hz, meeting the requirements for open-loop harvesting of stationary fruit, as validated in this work. This research conclusively demonstrates that accurate static perception forms the cornerstone of reliable robotic picking. Building on this foundation, future work will address current limitations through a structured plan: first, by prioritizing the development of advanced algorithms for localizing fruits under heavy foliage; second, by increasing the perception speed to enable real-time, closed-loop tracking of dynamically moving fruit; and finally, by integrating and validating these improvements within a complete robotic system for end-to-end dynamic harvesting.

CRediT authorship contribution statement

Shimin Hu: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation. **Meili Sun:** Data curation. **Chunjiang Zhao:** Supervision, Resources, Funding acquisition. **Ya Xiong:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Ya Xiong reports financial support was provided by the Haidian District Bureau of Agriculture and Rural Affairs. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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